

COMPLIANCE REPORT

Elliptic Bitcoin Transaction Network (Fraud Pattern Analysis)

To: Head of Compliance

From: KB (Data Analyst)

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Executive Summary

This report covers a network analysis of 203,769 Bitcoin transactions from the Elliptic dataset, testing for two laundering behaviours, layering (passing funds through multiple hands to obscure their origin) and fan-out (splitting flows into smaller transfers to avoid detection). Three transactions require urgent action. Eighteen unidentified transactions are displaying suspicious behaviour and should be escalated immediately.

Three Highest-Risk Transactions

Risk was assessed by combining four signals, incoming feeds, outgoing flows, centrality to payment routes, and influence from high-traffic nodes. The three transactions below scored highest across all four.

1. Transaction 111575373 (Risk Score: 0.636)

Class: licit

Pattern: Layering

Received funds from 13 separate sources before redistributing onward, the defining characteristic of layering. A licit label means it has not been flagged before, not that it is clean. It carries the highest risk score in the entire analysis.

2. Transaction 232438397 (Risk Score: 0.532)

Class: licit

Pattern: No pattern

Receives funds from over 160 distinct transactions, functioning as a consolidation point before funds move onward. This is consistent with the integration stage of laundering. If the account behind this transaction belongs to a legitimate business, it may be an unwitting vehicle.

3. Transaction 8856425 (Risk Score: 0.190)

Class: Unknown

Pattern: Layering and Fan-out.

The only transaction flagged by both detection methods simultaneously. It received from multiple sources and sent to three destinations within a two-step window, with no prior investigator review on record. This is the single highest-priority escalation in this report.

Most Suspicious Cluster

The analysis identified a cluster of 7,880 transactions with 9,164 payment flows, all active in the same early time window. Three patterns stand out. First, unidentified funds flow into licit class accounts at high volume,

over 45,000 flows, consistent with integration, the final stage of laundering where dirty money enters the legitimate financial system. Second, confirmed illicit transactions send more flows toward licit destinations than toward other illicit ones, suggesting a deliberate strategy to dilute traceability. Third, transaction 8856425 sits at the structural centre of this cluster, flagged by both detection methods, consolidating from multiple sources and dispersing rapidly, the equivalent of a pass-through node operating at speed. A cluster this connected and this active within a single time window warrants a full transaction trace.

Recommendations

Recommendation 1: Transaction Trace on Two High-Risk Nodes

Transactions 111575373 and 232438397 carry licit labels but display confirmed laundering patterns. The AML Investigations team should open a formal review of both within five working days, establish whether they map to known customer accounts, and determine whether a Suspicious Activity Report should be filed.

Recommendation 2: Label the 18 Flagged Unknown Transactions

Eighteen transactions flagged as suspicious carry no confirmed identity and cannot be escalated through existing workflows without one. The Data Governance team should obtain confirmed labels from the source data provider within 30 days. Until then, all 18 should be treated as high-risk in any screening or onboarding process.

Data Limitation

The working graph covers approximately 8,000 of the 203,769 transactions in the full dataset, deliberately focused on the most structurally connected part of the network. Transactions that appear on few paths and connect to few others fall outside this scope. The fact that none of the 4,545 confirmed illicit transactions appeared in the flagged results reflects this boundary, not an absence of fraud, confirmed illicit nodes in this dataset tend to sit at the edges of the network, suggesting deliberate peripheral routing. A follow-on analysis targeting known illicit nodes and their immediate neighbours is recommended to increase detection coverage.